MIL-R-51237A(EA)
24 August 1984
SUPERSEDING
MIL-R-51237(MU)
20 September 1965

MILITARY SPECIFICATION

REFILL KIT, CHEMICAL AGENT DETECTOR, V-G COMPONENTS, ABC-M30A1

This specification is approved for use by US Army Armament Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of refill kit for use with chemical agent detector kits. This refill kit includes anticholinesterase detector tickets with the necessary accessories, and a booklet of detector paper.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, US Army Armament, Munitions and Chemical Command, ATTN: AMSMC-TDC-S (A), Aberdeen Proving Ground, MD 21010-5423 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6665

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SPECIFICATIONS

FEDERAL

L-P-378 - Plastic Sheet and Strip, Thin Gauge, Polyolefin
NN-P-71 - Pallets, Material Handling, Wood, Stringer Construction, 2-Way and 4-Way (Partial)
UU-P-258 - Paper; Index
PPP-B-585 - Boxes, Wood, Wirebound
PPP-B-636 - Boxes, Shipping, Fiberboard
PPP-B-676 - Boxes, Setup
PPP-F-320 - Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes

MILITARY

MIL-P-116 - Preservation, Methods of
MIL-B-117 - Bags, Sleeves and Tubing - Interior Packaging
MIL-S-51189 - Substrate Solution and Dispenser

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

US ARMY ARMAMENT, MUNITIONS AND CHEMICAL COMMAND

CHEMICAL RESEARCH AND DEVELOPMENT CENTER

5-67-266 - Paper, Chemical Agent Detector, VGH, ABC-M8, Booklet of
5-77-808 - Belt, Detector Ticket
5-77-2100 - Bottle, Squeeze
5-77-2347 - Marking for Refill Kit, Chemical Agent Detector, V-G Components, ABC-M30A1

PUBLICATIONS

DEPARTMENT OF THE ARMY TECHNICAL MANUAL

TM 3-6665-254-12 - Detector Kit, Chemical Agent, ABC-M18A2

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(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

ASTM STANDARDS

D 3951 - Commercial Packaging

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Components. The refill kit shall contain one each of the items listed in table I in accordance with the applicable document.

TABLE I. Components of refill kit

Item	Applicable document
Booklet of ABC-M8 paper	Dwg 5-67-266
Belt of 40 detector tickets	Dwg 5-77-808
Bottle of tris-buffer	Dwg 5-77-2100-2
Substrate solution and dispenser	MIL-S-51189
Instruction card	See 3.2

3.2 Instruction card.

3.2.1 Material. The instruction card shall be made of white paper conforming to grade C of UU-P-258, 0.0105 inch thick and of minimum dimensions suitable to contain the necessary printed instructions.

3.2.2 Printing. The following instructions shall be printed in sans-serif type on the instruction card with black printer's ink.

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INSTRUCTIONS FOR USING THE REFILL KIT,

CHEMICAL AGENT DETECTOR, V-G COMPONENTS, ABC-M30A1

For use as a component of Refill Kits, Chemical Agent Detector, M18A2, see TM 3-6665-254-12, Detector Kit, Chemical Agent, ABC-M18A2.

3.3 Workmanship. The kit and components thereof shall be free from damage (rips, punctures, cracks, or tears) and contamination (dirt, grease, or oil).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the refill kits offered for acceptance at one time which have been produced by one manufacturer under essentially the same manufacturing conditions. No more than one batch of tris-buffer, substrate solution, detector tickets, or detector paper, shall be represented in one lot of refill kits.

4.2.2 Sampling.

4.2.2.1 For examination. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.3 Inspection procedure.

4.2.3.1 For examination. Sample refill kits and the packaging thereof shall be examined in accordance with the following classification of defects and MIL-STD-105.

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4.2.3.2 Classification of defects.

4.2.3.2.1 Refill kits.

Categories		Defects
Critical:		None defined
Major:	AQL 1.0 percent defective	
	101	Component missing
	102	Component incorrectly assembled
	103	Damage (see 3.3)
	104	Contamination (see 3.3)
	105	Marking missing or incorrect
	106	Marking illegible

4.2.3.2.2 Packaging (section 5).

Categories		Defects
Critical:		None defined
Major:	AQL 1.5 percent defective	
	101	Unit, intermediate, or shipping container not as specified
	102	Kit component missing from unit package
	103	Closure of unit, intermediate, or shipping container not as specified
	104	Unit, intermediate, or shipping container damaged
	105	Marking of unit, intermediate, or shipping container illegible, incorrect, or incomplete

5. PACKAGING

5.1 Preservation, level A.

5.1.1 Unit packing. One refill kit shall consist of a setup box conforming to type I, variety 2, class C, style 4 of PPP-B-676, which contains one each of the components listed in table I. The inside dimensions of the box shall be 4-3/4 by 3-3/8 by 1-5/8 inches (length, width and depth, respectively). The belt of detector tickets, prior to consolidating with the other table I components, shall be preserved in accordance with method IA-8 of MIL-P-116, using a bag conforming to type III, class B, style 2 of MIL-B-117, except that the bag shall be formed from polyethylene conforming to L-P-378.

5.1.2 Intermediate packing. Ten refill kits, unit packed as specified in 5.1.1, shall be intermediately packed in two layers of five unit packs per layer, within a fiberboard box conforming to class weather-resistant, grade W5c

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or W6c of PPP-B-636. The inside dimensions of the box shall be 18-1/4 by 5-1/4 by 4 inches (length, width and depth, respectively). Motion of contents in the box shall be prevented by inserting fiberboard pads where needed. The pads shall be formed from material conforming to class weather-resistant, grade W5c or W6c of PPP-F-320. The box shall be closed in accordance with the requirements for closure method IV of PPP-B-636.

5.2 Packing. Refill kits shall be packed level A, B or industrial as specified (see 6.2).

5.2.1 Level A. Two intermediate packs of refill kits of 5.1.2 shall be packed in a wirebound wood box conforming to class 3, style optional for a type 2 maximum load of 125 pounds of PPP-B-585. The inside dimensions of the box shall be 18-3/4 by 11-1/2 by 4-5/8 inches (length, width and depth, respectively). Motion of contents in the box shall be prevented by inserting fiberboard pads, where needed, formed from material conforming to class weather-resistant, grade V3c of PPP-B-636. The box shall be closed as specified in PPP-B-585.

5.2.2 Level B. Two intermediate packs of refill kits of 5.1.2 shall be packed in the manner as specified in 5.2.1 for level A except that the box shall conform to class weather-resistant, grade V3c of PPP-B-636, and shall be reinforced after closure as specified in PPP-B-636.

5.2.3 Industrial. Two intermediate packs of refill kits of 5.1.2 shall be packed in accordance with ASTM D 3951.

5.2.4 Unitization. Level A and B packs of refill kits shall be palletized in accordance with load type I of MIL-STD-147, using the soft wood pallet conforming to type IV of NN-P-71. Industrial packs shall be unitized in a manner to protect their contents, assure acceptance by common carrier and stability in transportation and storage for a minimum duration of six months.

5.3 Marking. Level A unit and intermediate packs shall be marked in accordance with MIL-STD-129. Level A and B packs shall be marked as shown on Drawing 5-77-2347. Industrial packs shall be marked as specified in ASTM D 3951.

5.3.1 Special marking. Unit and intermediate packs and industrial packs shall be marked to show a refill kit discard date as follows: "DISCARD DATE [ ]/[ ]." The symbols "[ ]" represent the month and year of discarding. For example "09/94" represents a discard date of September 1994. The discard date shall be 3 years from the date of manufacture of the refill kit.

6. NOTES

6.1 Intended use. This refill kit is intended for detector kit users that only have need for the detector tickets with the necessary accessories and the booklet of M8 paper, when they are exhausted.

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6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification, and
(b) Level of packaging required.

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 6665-A454

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-R-51237A(EA)	REFILL KIT, CHEMICAL AGENT DETECTOR, V-G COMPONENTS, ABC-M30A1

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation.

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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